

Nieb Hasan Neom

Computer Science PhD Student | AI Systems for Distributed Systems & Networking | Backend Engineering

CONTACT

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FOCUS

AI systems
High-performance networking
Backend systems
Reproducible evaluation

PROBLEM SOLVING

Codeforces Expert
Reached Expert with a peak rating of 1629.

LeetCode Knight
Reached Knight with a peak contest rating of 1895.

KonaSL Coding Combat 2023
Placed first runner-up in a 20-team field.

SUMMARY

I'm Nieb Hasan Neom, a Computer Science PhD student and Kummer Doctoral Fellow building AI systems for distributed systems and networking. My work combines guarded LLM orchestration and agent-driven data movement with three years of backend engineering.

TECHNICAL SKILLS

AI & Agentic Systems: LLM function calling and tool use · Schema-constrained generation · Agentic workflow orchestration · Ollama and Hugging Face models · PyTorch, Pydantic, and NumPy

Systems & Networking: Python, C, C++, SQL, and Bash · TCP performance analysis · Concurrency tuning and multiprocessing · Zero-copy sendfile() and Linux systems programming · CloudLab, FABRIC, and TACC/ACCESS

Distributed Systems & Backend: Java and Spring Boot · Apache Kafka and message queues · Microservice and REST API design · AWS S3, SNS, and EC2 · PostgreSQL, MongoDB, Redis, and MySQL

Engineering Tooling: Git and Gradle · Jenkins and Jira · IntelliJ IDEA, DataGrip, and VS Code · LaTeX · OpenAI and Google Gemini APIs

Graduate Research Assistant · Missouri University of Science and Technology

August 2025 – Present

Researching machine learning for distributed systems and networking in the Computer Systems & Networking Lab with Dr. Md Arifuzzaman. Current work covers guarded scientific-model orchestration and bounded, telemetry-driven data-movement optimization.

Software Engineer II · Kona Software Lab Ltd.

April 2022 – June 2025

As a backend engineer on a blockchain-based token-trading platform, I worked on trade, user, wallet, and fee-management services; introduced message queues for reliable inter-service communication; added caching to reduce response latency; and implemented two-factor authentication for sensitive trading operations.

EDUCATION

Doctor of Philosophy in Computer Science

August 2025 – Present

Missouri University of Science and Technology

Bachelor of Science in Computer Science and Engineering

February 2017 – April 2022

Khulna University of Engineering and Technology

SELECTED WORK

Guarded Scientific Model Orchestration

A state-aware orchestration layer that extends LLM function calling with explicit workflow semantics so domain scientists can invoke trained prediction models through constrained natural language.

Kona Token Trade

Backend service work for a blockchain-based token-trading platform, centered on reliable service communication, latency-conscious caching, and stronger controls for sensitive operations.

Bangla Sign Language Recognition

Undergraduate thesis research that contributed to a published hybrid recognition system combining image preprocessing, transfer learning, and a random-forest classifier.

Agentic Data Transfer Optimizer · Research in Progress

A bounded agentic system that adapts network-transfer concurrency from live TCP telemetry while making every recommendation, constraint, and state transition inspectable.

PUBLICATIONS

Guarded, State-Aware Orchestration for Schema-Constrained Scientific Model Execution

2026 · accepted · 21st Workshop on Workflows in Support of Large-Scale Science (WORKS26 at SC26)

Self-Tuning Segmented Retrieval of Large-Scale Genomic Data

2026 · accepted · 22nd IEEE International Conference on e-Science (eScience 2026) · Short paper

A hybrid approach for Bangla sign language recognition using deep transfer learning model with random forest classifier

2023 · published · Expert Systems with Applications 213, article 118914

A Review on Fall Detection Systems Using Data from Smartphone Sensors

2019 · published · Ingénierie des Systèmes d'Information 24(6), 569–576

CredSec: A Blockchain-based Secure Credential Management System for University Adoption

2024 · preprint · arXiv:2406.05151 · Preprint

LEADERSHIP & COMMUNITY

Programming education

Led an introductory competitive-programming workshop for more than 60 KUET students and helped organize programming contests.

KUET Math Club

Served as Assistant Organizing Secretary and helped run math events and Olympiad training.

Sports leadership

Organized university sports activities and captained the CSE cricket team.